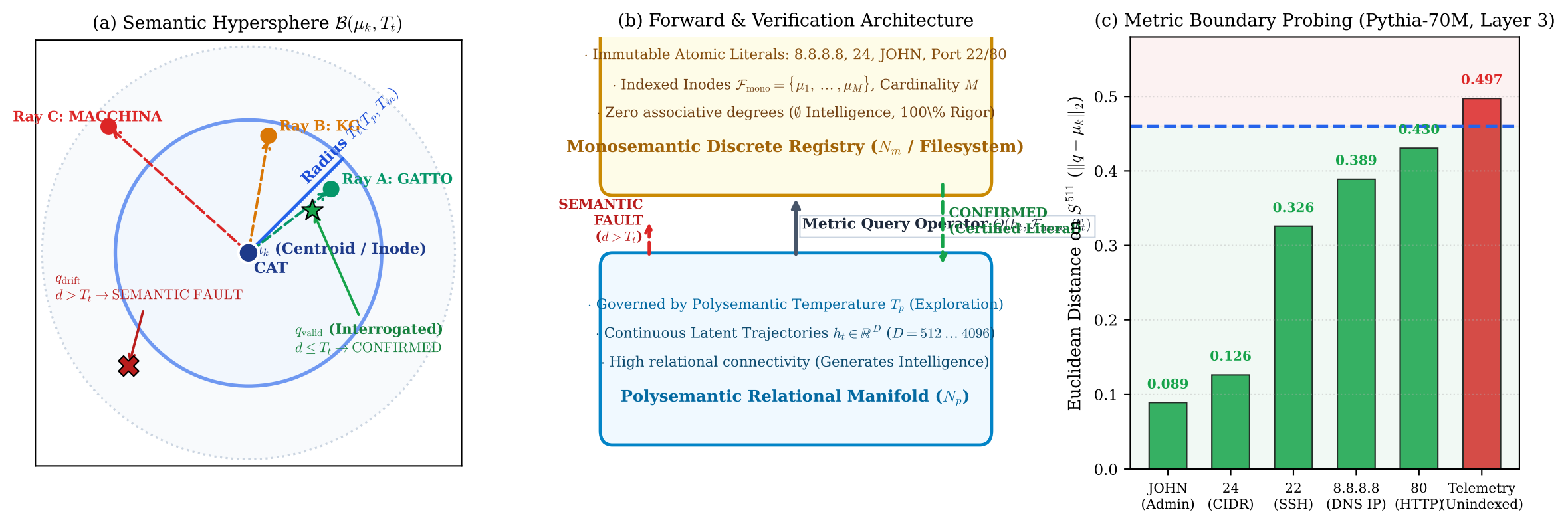

(a) Semantic Hypersphere $\mathcal{B}(\mu_k, T_t)$
Ray C: MACCHINA
Ray B: KG
Radius $T_t(T_p, T_m)$
Ray A: GATTO
ι_k (Centroid / Inode)
CAT
q_{drift}
$d > T_t \rightarrow$ SEMANTIC FAULT
q_{valid} (Interrogated)
$d \leq T_t \rightarrow$ CONFIRMED
(b) Forward & Verification Architecture
· Immutable Atomic Literals: 8.8.8.8, 24, JOHN, Port 22/80
· Indexed Inodes $\mathcal{F}_{\text{mono}} = \{\mu_1, \ldots, \mu_M\}$, Cardinality M
· Zero associative degrees ($\emptyset$ Intelligence, 100\% Rigor)
Monosemantic Discrete Registry (N_m / Filesystem)
SEMANTIC FAULT ($d > T_t$)
Metric Query Operator $\mathcal{O}(h_t, \mathcal{F}_{\text{mono}})$
CONFIRMED (Certified Literal)
· Governed by Polysemantic Temperature T_p (Exploration)
· Continuous Latent Trajectories $h_t \in \mathbb{R}^D$ ($D = 512 \ldots 4096$)
· High relational connectivity (Generates Intelligence)
Polysemantic Relational Manifold (N_p)
(c) Metric Boundary Probing (Pythia-70M, Layer 3)
Euclidean Distance on S^{511} ($\|q - \mu_k\|_2$)
0.0
0.1
0.2
0.3
0.4
0.5
0.089
0.126
0.326
0.389
0.430
0.497
JOHN (Admin)
24 (CIDR)
22 (SSH)
8.8.8.8 (DNS IP)
80 (HTTP)
Telemetry (Unindexed)